

activity in sub 100 share quantities....Allowing for access to this information, as proposed, would therefore likely result in increased pre-trade transparency for both retail and institutional investors and bolster execution quality.”¹⁶⁰¹

In addition, the amendments will change the timing and magnitude of compliance costs and other costs.¹⁶⁰² One commenter estimated that quotation traffic will increase at least 35% as a result of adding odd-lot data to the SIP feeds; this estimate was based on a previous proposal by the CTA and UTP Operating Committees, which proposed a more limited inclusion of odd-lot data to the SIP feeds.¹⁶⁰³ The associated costs will include: the cost for exclusive SIPs to upgrade existing infrastructure and software to handle the dissemination of additional message traffic,¹⁶⁰⁴ the cost to SROs to implement system changes required in order to make the data needed to generate odd-lot information available to exclusive SIPs, and the cost of technological investments market participants might have to make in order to receive the SIP data.¹⁶⁰⁵

While these aforementioned economic effects of including odd-lots in NMS data will be realized sooner, the Commission does not expect that the amendments will accelerate all the effects described in the MDI Rules related to adding to NMS data odd-lot information inside the NBBO. The amendments will not accelerate the decentralized consolidation model and will therefore not accelerate the benefits from allowing some market participants to reduce data

¹⁶⁰¹ See BlackRock Letter at 11.

¹⁶⁰² See MDI Adopting Release, supra note 10, at 18759 for the full discussion of the costs associated with expanding core data to include odd-lot information inside the NBBO. See also infra section VII.D.5.c for further discussion of compliance costs.

¹⁶⁰³ See FISD Letter at 3. The Commission agrees that the addition of information on odd-lot quotes that are priced at or more aggressively than the NBBO may substantially increase message traffic. See MDI Adopting Release, supra note 10, at n.2019.

¹⁶⁰⁴ Multiple commenters agreed with the amendments’ effect on message traffic. See, e.g., Citadel Letter I at 26 and FIA PTG Letter II at 4.

¹⁶⁰⁵ See supra note 1602.

expenses required for trading by providing a reasonable alternative to some market participants to proprietary data.¹⁶⁰⁶ As such, the amendments will also not accelerate the cost to users of proprietary data whose information advantage will dissipate somewhat. In particular, the Commission does not believe that adding the specified odd-lot information to the exclusive SIPs will result in low-latency traders substituting the exclusive SIPs for their current proprietary data usage. This is because a key component of the MDI Rules for this functionality is an expected reduction in latency of NMS data anticipated from the competing consolidator model of NMS data distribution.¹⁶⁰⁷ The exclusive SIPs are not expected to be fast enough to replace proprietary data because existing SIP latency will not be reduced or affected by this Rule. Thus, the amendments will not accelerate the benefits anticipated in the MDI Rules that pertain to using low-latency odd-lot information. Instead, the Commission expects these effects to be realized after the implementation of all MDI Rules.

Market participants who decide to receive and use odd-lot quotation information from the exclusive SIPs under these amendments will also incur costs if the acceleration results in additional systems changes when competing consolidators begin offering odd-lot information. Specifically, market participants that decide to receive odd-lot quotation information from exclusive SIPs will need to make systems changes upon implementation of the acceleration of the MDI Rules in order to receive the odd-lot quotation information. Because the data specifications of the competing consolidators are unknown and could differ from the data specification of the exclusive SIPs, market participants receiving odd-lot information from the exclusive SIPs could also need to make systems changes again to receive the odd-lot information

¹⁶⁰⁶ Id.

¹⁶⁰⁷ See MDI Adopting Release, supra note 10, at 18752 n.1939.

from a competing consolidator upon full implementation of the MDI Rules.¹⁶⁰⁸ If there are significant fixed costs associated with system changes that are incurred on each change, then multiple system changes will be inefficient and could increase costs. Because market participants who receive odd-lot quotation information from the exclusive SIPs may need to make an extra systems change stemming from this Rule—one change to receive the data from the exclusive SIPs, and potentially another change to receive the data from a competing consolidator—some market participants may decide not to implement systems changes to make use of the accelerated implementation of the odd-lot information and, instead, wait until the MDI Rules are fully implemented. This would dampen some of the benefits of accelerating the inclusion of odd-lot quotation information.

To the extent that some market participants store SIP data for various purposes (such as transaction cost analysis), the acceleration of the MDI Rules could hasten an increase in storage costs because the amount of SIP data increases with the inclusion of odd-lot data. Many factors affect these costs in total, such as the number of market participants storing SIP data, the data structures they use to store SIP data, whether these market participants will choose to store all or just some of the SIP data provided by the amendments, and the period over which the amendments will affect these storage costs. Because the Commission does not have information on how many market participants will store MDI odd-lot information and the methods they will use to do so, the Commission is unable to estimate these costs.

¹⁶⁰⁸ One commenter pointed out that this duplication of effort becomes more likely if SIPs do not choose to register as competing consolidators. See BlackRock Letter at 12. The Commission agrees with this assessment.

One commenter stated that displaying odd-lot quotes “could lead investors to expect prices that are not available.”¹⁶⁰⁹ The Commission acknowledges that there may be a learning curve associated with the dissemination of odd-lots on the SIP. However, any confusion from the accelerated dissemination of odd-lot quotes would have occurred eventually under the original MDI timeline, so the incremental effect of MDI acceleration on investor confusion is minimal. Further, as discussed in the MDI Adoption Release, the Commission acknowledges that many retail investors may not directly view the entire content of expanded core data—retail brokers may decide not to offer their customers direct access to all of the odd-lot information but may rather customize products derived from odd-lot information.¹⁶¹⁰ The provider of these customized products is expected to supply the information in a way that does not confuse the provider’s customers. To the extent that retail brokers allow some customers to directly utilize all of the odd-lot information, the customers who choose to do so will likely be sophisticated—as evidenced by their seeking out the information—and will likely not be confused.¹⁶¹¹

c. Dissemination of Odd-Lots in SIP Data

The Amendments require the exclusive SIPs to disseminate odd-lot data.¹⁶¹² As discussed in the Proposing Release,¹⁶¹³ this requirement will help realize the benefits of accelerating the implementation of including odd-lot information in NMS data while imposing costs on exclusive SIPs and potentially on market participants.¹⁶¹⁴ The MDI Rules do not require the competing

¹⁶⁰⁹ See Schwab Letter II at 36.

¹⁶¹⁰ See MDI Adopting Release, *supra* note 10, at 18753.

¹⁶¹¹ See MDI Adopting Release, *supra* note 10, at 18754.

¹⁶¹² See 17 CFR 242.603(b)(3) for rule text relating to this requirement under Rule 603.

¹⁶¹³ See Proposing Release, *supra* note 11, at 80332.

¹⁶¹⁴ See *infra* section VII.D.5.c for additional discussion of the costs the exclusive SIPs are expected to incur.

consolidators to disseminate odd-lot data. However, the Commission estimated in the MDI Rules that at least one competing consolidator will do so because there will be demand for the data.¹⁶¹⁵ These amendments, though, do not accelerate the competing consolidator model. Unlike competing consolidators, each exclusive SIP is the only distributor of the entirety of its data and may lack the incentive to disseminate the data. As a result, the Commission cannot rely on the exclusive SIPs to disseminate the odd-lot information prescribed by the MDI Rules absent a requirement to do so; the benefits of the acceleration could therefore be at risk without the Amendment to Rule 603's requirement for the SIPs to disseminate.¹⁶¹⁶

While the inclusion of the odd-lot data could impose costs on those who receive and use exclusive SIP odd-lot data,¹⁶¹⁷ the requirement that exclusive SIPs disseminate the data could also impose costs on those who receive but do not have an interest in using odd-lot information provided in SIP data. These costs would vary based on how the exclusive SIPs decide to implement the dissemination of the MDI odd-lot information. For example, if the exclusive SIPs offer a separate data feed for odd-lot quotation information, then market participants that do not have an interest in this information may not incur any additional costs because they would not need to subscribe to this data feed. If the exclusive SIPs instead incorporate MDI odd-lot quotation information into an existing data feed, then market participants may incur costs to update their systems to filter out the unwanted odd-lot information; the Commission is unable to estimate these costs because they would vary across market participants and depend upon each

¹⁶¹⁵ See supra note 1155.

¹⁶¹⁶ The Commission recognizes that the exclusive SIPs have some incentive to offer odd-lots as indicated by the exclusive SIPs seeking comment on doing so. See, e.g., Proposal of the CTA and UTP Operating Committees Regarding Odd Lots on the SIPs (Mar. 2022), available at https://www.ctaplan.com/publicdocs/ctaplan/CTA_Odd_Lots_Proposal_2022.pdf.

¹⁶¹⁷ These costs include systems changes and data storage. See section VII.D.4.b for a discussion of these costs.

market participant's existing infrastructure, which is unknown to the Commission. Further, such SIP data users could incur the cost of any SIP data fee increases intended to offset the costs to exchanges and exclusive SIPs.¹⁶¹⁸ However, SIP data fees did not increase when the exclusive SIPs started to include odd-lot trades.

d. Best Odd-Lot Order Definition

The amendments go beyond the MDI Rules by requiring that NMS data also include information on the best priced odd-lot orders across all markets. Including the best odd-lot order in a standardized form will offer market participants a standard benchmark, like the NBBO, to use to measure execution quality. As discussed in the Rule 605 Amendments¹⁶¹⁹ a market center may be able to internalize an order and claim price improvement relative to the NBBO even if better priced odd-lots are available at another market center. A standardized best odd-lot benchmark may give market participants that receive it valuable information for evaluating broker-dealers and market centers. Including this benchmark in NMS data allows the information to be readily available to a broad set of market participants, including those investors to whom broker-dealers choose to make this information available.¹⁶²⁰

¹⁶¹⁸ Any changes in fees for SIP data would need to be filed by the Equity Data Plans and approved by the Commission. See supra note 887 and accompanying text for further discussion.

¹⁶¹⁹ See Rule 605 Amendments, supra note 10, section IX.D.b.2.c.vii for further discussion of the benefits of disclosing execution quality benchmarked to the best available displayed price.

¹⁶²⁰ Because the amount of information disseminated as a result of the amended rule BOLO dissemination requirement would likely result in significantly less message traffic compared to the amount of information that would be disseminated as a result of the amended rule MDI odd-lot information dissemination requirement, we expect market participants will be able to receive the BOLO information required here without having to make significant system upgrades, unlike the MDI odd-lot information. Therefore, a broader set of market participants may choose to receive the BOLO from the exclusive SIPs (and later on, competing consolidators) than all of the MDI odd-lot information. Additionally, because of the lower message traffic, more broker-dealers may make information on the BOLO available to their customers than MDI odd-lot information.

Currently, this information is only available to market participants who have proprietary data feeds, and even then there could be differences across market participants with these data in terms of how exactly market participants calculate the best odd-lot order (or how many proprietary feeds they include). The best odd-lot information in the NMS data will provide a standardized benchmark that reflects the best odd-lot price consolidated across all national securities exchanges and national securities associations. This benchmark may allow more market participants to better monitor the execution quality of their broker-dealers and send more trading volume to broker-dealers with better performance.¹⁶²¹ One commenter highlighted the value of this new benchmark: “...transparency requires that the units used to represent the range of prices available in the market match the units in which participants typically quote and trade.”¹⁶²² For market participants who receive the BOLO and typically execute small trades, the best odd-lot order will provide them more relevant information on available orders. Thus, including the best odd-lot information could enhance competition among broker-dealers leading to better trade execution and perhaps a lower cost to customers for execution services.

One commenter expressed concern that the best odd-lot order would result in “displaying locked/crossed markets.”¹⁶²³ It is possible for the best odd-lot bid to be at a price equal to or higher than the best odd-lot ask; in these cases, the best odd-lot order would show a locked or

¹⁶²¹ While the Commission does not expect most retail traders would engage in this sort of benchmarking due to a lack of technical capacity to do so among most retail traders, institutional traders likely have such capacity and so would engage in this type of monitoring. Institutional traders have strong incentives to monitor all aspects of transaction costs as these costs can significantly affect portfolio performance. See Amber Anand, et al., Performance of Institutional Trading Desks: An Analysis of Persistence in Trading Costs, 25 REV. FIN. STUD. 557 (2012).

¹⁶²² See IEX Letter I at 31. Another commenter agreed more generally that the best odd-lot order will enhance the usefulness of odd-lot information and enhance liquidity—see Cboe Letter II at 10.

¹⁶²³ See SIFMA Letter II at 34 requesting an analysis of the effect of the BOLO on the display of locked and crossed markets.

crossed market.¹⁶²⁴ Academic research shows that the NBBO does get locked and crossed from time to time.¹⁶²⁵ These tend to be fleeting events.¹⁶²⁶ Because the BOLO will provide prices inside the NBBO, the BOLO will likely be crossed or locked more frequently than the NBBO. However, it is unclear what practical effect a locked or crossed BOLO would have on financial markets or those that use the BOLO. Market participants are already well versed in using the NBBO, which can be locked and crossed from time to time, so it is likely that they would use similar techniques for dealing with locked and crossed markets when, for example, benchmarking relative to the BOLO.¹⁶²⁷ Further, market participants who subscribe to proprietary data feeds already have access to information on when the best odd-lot orders may lock or cross each other; the rule amendment merely extends this information to market participants who do not subscribe to proprietary data feeds. As more market participants see the information contained in the BOLO, there may be fewer instances of locked and crossed odd-lot quotes—e.g., more market participants will have the information needed to arbitrage crossed odd-lot markets. Finally, a locked or crossed BOLO will be less disruptive than a locked or crossed NBBO because Rule 610 of Regulation NMS requires SROs to adopt rules requiring their members reasonably to avoid displaying quotations that lock or cross protected

¹⁶²⁴ See supra note 1051 for the definition of locked and crossed markets. A locked or crossed market occurs when there is a passive buy order on one venue at a price greater or equal to the price of an existing passive sell order at another venue; the fact that these orders have not executed against each other indicates that there is a friction between the trading venues.

¹⁶²⁵ See Craig W. Holden & Stacey Jacobsen, Liquidity Measurement Problems in Fast, Competitive Markets: Expensive and Cheap Solutions, 69 J. FIN. 1747 (2014). This paper estimates that 1.7% of trades occur when the NBBO is locked, and 0.5% of trades occur when the NBBO is crossed—see Table 1, Panel A, column 4 therein. The authors conjecture that some of these instances arise due to a data issue where quotes have been canceled, but the cancellation was not recorded by the time of the trade.

¹⁶²⁶ Id.

¹⁶²⁷ See Id. for an example of one methodology used when employing market data in the presence of locked or crossed markets.

quotations.¹⁶²⁸ However, the BOLO does not establish a protected quote, and so a locked or crossed BOLO would not trigger the same reaction by SROs and their members as a locked or crossed NBBO.

Other commenters discussed the effect BOLO may have on investor confusion. Two commenters stated that: “Calculating and publishing an odd-lot NBBO risks creating significant investor confusion due to the appearance that a new benchmark is being established even though odd-lots are treated differently than round-lots under Commission regulations. Rather than taking steps to prevent unnecessary investor confusion, the Commission encourages it by suggesting that the odd-lot NBBO is a ‘standard benchmark’ that could be used by investors ‘to measure the amount of price improvement they receive for the execution of their orders.’”¹⁶²⁹ The commenters continued: “The odd-lot NBBO is not a standard benchmark, since the size associated with these quotes will vary greatly as opposed to the actual NBBO, which always represents a round-lot.”¹⁶³⁰ Similarly, some commenters stated that the BOLO will not provide a useful benchmark and may instead distort price improvement statistics.¹⁶³¹

The Commission acknowledges that the BOLO is not, at present, a widely used and standard benchmark. This is likely because the requisite data is not broadly distributed and is only available to market participants who have proprietary data feeds. In contrast, the NBBO is broadly distributed in NMS data and is also more widely used as a benchmark. The Commission

¹⁶²⁸ 17 CFR 242.610(d)(1).

¹⁶²⁹ See Citadel Letter I at 26-27, and SIFMA Letter II at 43-44.

¹⁶³⁰ Id.

¹⁶³¹ See JPMorgan Letter at 7, FIA PTG Letter II at 5, and ASA Letter at 6.

believes that including the BOLO in NMS data will similarly allow market participants to more easily use the BOLO as a benchmark if they choose to.¹⁶³²

The Commission also recognizes that an odd-lot price that is better than the NBBO may not reflect sufficient quantity to execute certain orders, particularly larger-sized orders, and, as a result, price improvement relative to the BOLO will be more relevant in some cases than for others. However, market participants are already well versed in interpreting nuanced benchmarks—for example, holding the round lot size constant, the NBBO may reflect a different amount of dollar liquidity based on the price of the stock. Furthermore, the size available at the NBBO of any particular stock might be a great deal more than a single round lot. This means that market participants already deal with the distinction between the price of a benchmark and the amount of shares available at that benchmark price. Indeed, while one commenter points out the challenge of comparing a 500-share order to a 10-share odd-lot,¹⁶³³ a similar challenge already exists in comparing a 1000-share order to a 100-share round lot. This challenge is understood and handled already; market participants already know that quantity must be taken into account when making comparisons. The Commission expects that market participants who will benchmark their trades with the BOLO will generally have comparable levels of sophistication as investors who currently use the NBBO benchmark; given that users of the NBBO benchmark are already adept at accounting for order size, these users should not be confused by the BOLO. It is important that market participants have access to a variety of benchmarks to meet their

¹⁶³² See infra section VII.D.6.a.ii discussing how the BOLO will make it easier for market centers and broker-dealers to compute statistics on price improvement relative to the best available displayed price, now required by amended Rule 605.

¹⁶³³ See Citadel Letter I at 27.